The empirical recurrence for OEIS A223687, proved by a two-line transfer matrix

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Abstract

OEIS A223687 counts arrays over $\{0,\ldots,15\}$ in which every cell is constrained by the values of its neighbours, and carries an empirical recurrence of order 2 contributed by R. H. Hardin. It is true, and it is not an empirical matter. The neighbourhood reaches one row up and one row down, so a single row is not enough state and a plain walk count is wrong; the right object is a walk on ordered *pairs* of consecutive rows, where each step tests the condition on the middle row against both its neighbours and the two ends are tested with one neighbour row absent. That makes $a(n)$ a walk count on a digraph with $S = 4096$ vertices, and the conjectured recurrence is then decided by a finite exact computation.

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1 The sequence and the conjecture

OEIS A223687 is "Petersen graph (8,2) coloring a rectangular array: number of n X 3 0..15 arrays where 0..15 label nodes of a graph with edges 0,1 0,8 8,14 8,10 1,2 1,9 9,15 9,11 2,3 2,10 10,12 3,4 3,11 11,13 4,5 4,12 12,14 5,6 5,13 13,15 6,7 6,14 7,0 7,15 and every array movement to a horizontal or antidiagonal neighbor moves along an edge of this graph.". It has offset 1 and begins

$$144, 2304, 37008, 595584, 9594000, 154616832, 2492365968, 40180445568, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 24*a(n-1) - 127*a(n-2).

As of the "Last modified" line on the live entry (Aug 22 2018 09:52:15, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 24\,a(n-1) - 127\,a(n-2) \tag{1}$$

for all $n > 2$.

2 The condition, and why one row is not enough

The arrays have 3 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u, each in $\{0,\ldots,15\}$. With the neighbour offsets

$$\mathcal{N} = \{(0,1),\ (1,-1)\},$$

written as (row shift, column shift) and counting only positions inside the array, the entry's requirement is

$$\{x_{t,u}, x_{t+d_1,u+d_2}\} \in E \text{ for every offset in } \mathcal{N}, \quad E = \{\{0,1\},\{0,7\},\{0,8\},\{1,2\},\{1,9\},\{2,3\},\{2,10\},\{3,4\},\{3, \tag{2}$$

for every cell (t,u).

Some offsets in $\mathcal{N}$ have a positive row shift and some a negative one, so the condition at a cell of row t involves rows $t-1$, t and $t+1$ together. A transfer matrix whose states are single rows cannot express it: the step from $r^{(t)}$ to $r^{(t+1)}$ does not know $r^{(t-1)}$. Two consecutive rows are enough, and that is the state used below.

3 The digraph

Let $V = \{0,\dots,15\}^3$ be the possible rows and take $\mathcal{V} = V \times V$, of size $S = 4096$, as the vertex set. Declare $(p,c) \to (c,x)$ an edge when every cell of the middle row c satisfies (2) with p above it and x below it; there is no edge between pairs that do not overlap in this way. Let N be the adjacency matrix, $\mathbf{v}$ the indicator of the pairs (p,c) whose first row p satisfies (2) with no row above, and $\mathbf{u}$ the indicator of the pairs (p,c) whose second row c satisfies it with no row below.

Lemma 1. *For $L \geq 2$, the arrays counted by the entry with L rows are exactly the walks*

$$(r^{(1)}, r^{(2)}) \to (r^{(2)}, r^{(3)}) \to \cdots \to (r^{(L-1)}, r^{(L)})$$

of length $L-2$ starting at a vertex marked by $\mathbf{v}$ and ending at one marked by $\mathbf{u}$. Consequently

$$a(n) \;=\; \mathbf{v}^{\top} N^{L-2}\mathbf{u}, \qquad L = n.$$

Proof. Consecutive vertices of such a walk overlap in one row, so a walk of that shape is precisely a sequence $r^{(1)},\dots,r^{(L)}$ of rows, that is, an array of L rows and 3 columns. In the walk, the cells of row t for $2 \leq t \leq L-1$ are tested exactly once — at the step leaving $(r^{(t-1)}, r^{(t)})$ — and with their true neighbours $r^{(t-1)}$ and $r^{(t+1)}$. The cells of row 1 are tested by $\mathbf{v}$ and those of row L by $\mathbf{u}$, in both cases by the same condition with the missing neighbour row treated as absent, which is how the array itself behaves at its boundary. So the walks are exactly the admissible arrays, and summing over all starting and ending vertices counts them. □

The size of $\mathcal{V}$ is worth seeing concretely. There are $15+1 = 16$ values and 3 cells in a row, so $|V| = 16^3 = 4096$, and the vertex set is $V \times V$, of size $4096^2 = 4096$. That is the whole of the state: two consecutive rows and nothing else about the array's history.

It is worth saying why this is not done by enumeration. At the largest index the entry publishes, $n = 15$, the arrays have $3 \cdot 15$ cells over 16 values, so there are about 10^{54} of them to sift. The walk count replaces that by 4096 states and one matrix–vector product per row.

4 The criterion

Let $q(t) = t^2 - \sum_i c_i\, t^{2-i}$ be the characteristic polynomial of (1) and put $G = N$, so that a unit step in n advances the walk by 1 row.

Lemma 2. *Let n_0 be the least index with $L \geq 2$ and $h_j = G^j N^o \mathbf{u}$, where $o = 1n_0 + 0 - 2$. Then for $j \geq 2$,*

$$a(n_0+j) - \sum_i c_i\, a(n_0+j-i) \;=\; \mathbf{v}^{\top} G^{j-2} w, \qquad w = h_2 - \sum_i c_i\, h_{2-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^{\top} G^m w = 0$ for every $m \geq 0$, and it is enough to check $m = 0, 1, \dots, S-1$.

Proof. Substitute Lemma 1 and factor G^{j-2} out of $G^j - \sum_i c_i G^{j-i}$; the nonzero scalar 1 does not affect vanishing. For the last clause, Cayley–Hamilton makes G^S an integer combination of $I, G, \dots, G^{S-1}$, so $\mathbf{v}^\top G^m w$ for $m \geq S$ is the corresponding combination of the earlier values; if those vanish, so do all the rest. $\square$

Theorem 3. *The recurrence* (1) *holds for every $n > 2$, which is the statement of Section 1.*

Proof. The digraph was built directly from (2): for each of the $S = 4096$ pairs and each of the $(15+1)^3$ possible next rows, the condition was tested on every cell of the middle row. The vector w was formed by 2 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 2 + 1$ onwards, and the run was continued until $S + 1$ consecutive zeros had been seen, which by Lemma 2 settles every larger m at once. $\square$

Remark 4. The argument gives more than the recurrence: the sequence is C-finite with characteristic polynomial dividing that of G, hence has a rational generating function whose denominator divides $\det(I - xG)$. The conjectured recurrence is one consequence; the minimal one is a divisor of q.

5 Verification

Three checks, each independent of the algebra above.

First, the walk counts of Lemma 1 were compared against the entry's DATA at every one of the 15 published indices, with no shift fitted: the number of rows is read off the entry's own name as $L = n$ and the entry's offset says which index the first published term carries. The values agree exactly. A misreading of (2), of the neighbour set, or of the boundary treatment would give a different digraph and different counts, so this is a test of the modelling and not only of the arithmetic.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix, and returned zero throughout.

References

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